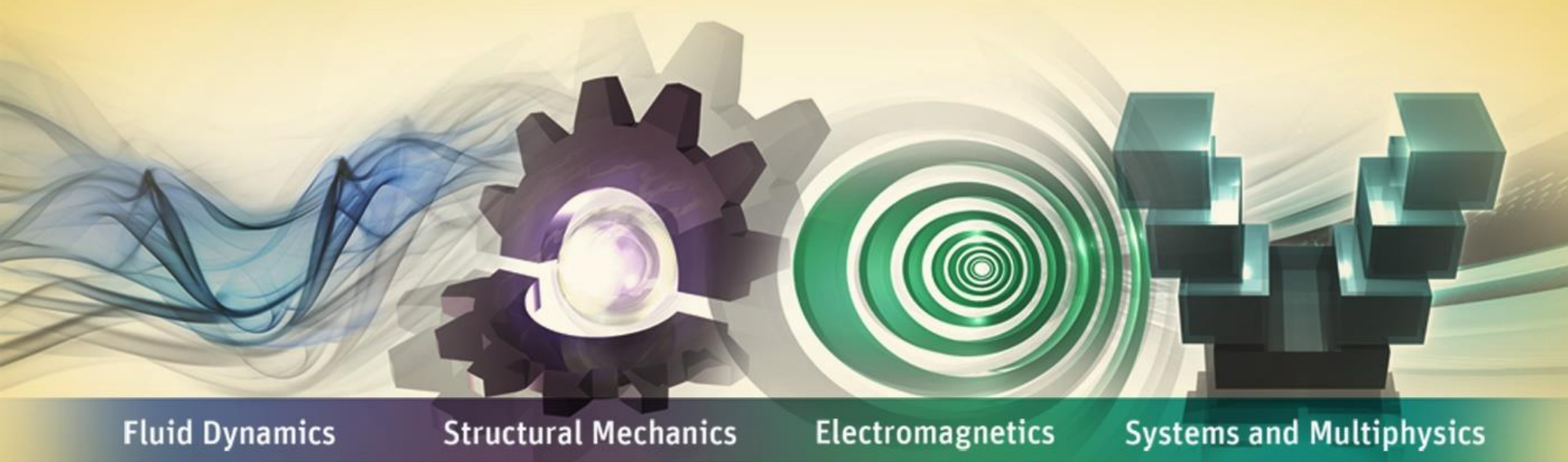


Workshop 6A: Contact Diagnostics



ANSYS Mechanical Introduction to Structural Nonlinearities

Workshop 6A: Diagnostics

Goal

Diagnose convergence trouble with surface body contact model

Model Description

- 3D Spring plate –Surface Body
- 3D Rigid Target Body
- Linear steel material
- Meshed with 3D SHELL elements
- Spring Fixed support at one end, A
- Rigid Body displaced into Spring 19mm

A: Static Structural (ANSYS)

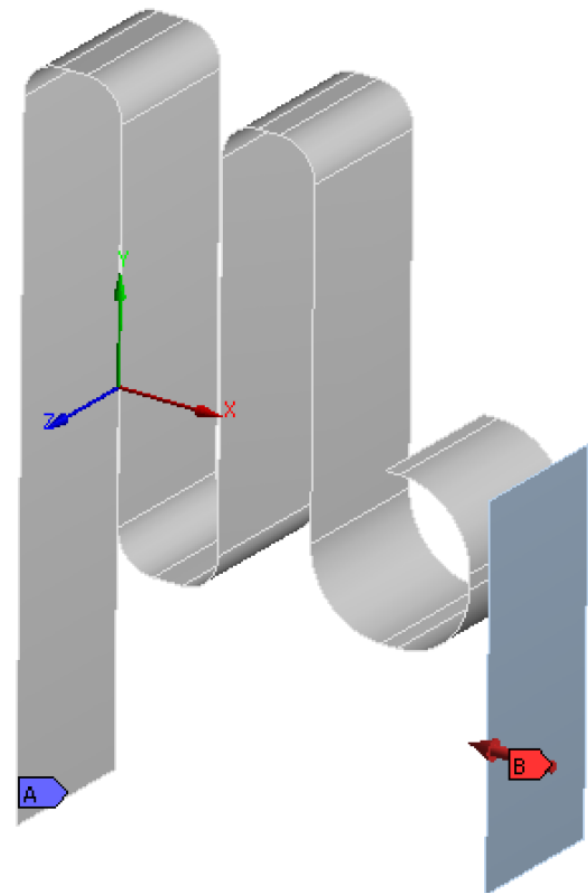
Displacement

Time: 1. s

3/5/2009 9:26 AM

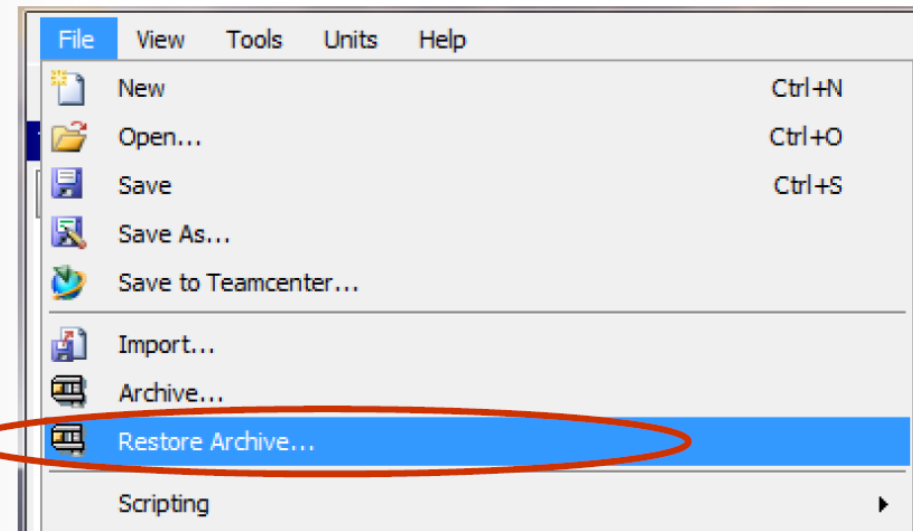
A Fixed Support

B Displacement



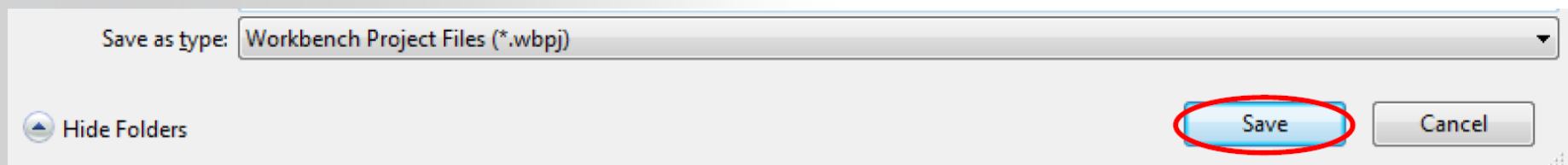
Steps to Follow:

Restore Archive... browse for file “SNL WS6a-diagnostics.wbpz”



Save as

- **File name: “WS6a-diagnostics”**
- **Save as type: Workbench Project Files (*.wbpj)**



...Workshop 6A: Diagnostics

The project Schematic should look like the picture to the right.

- From this Schematic, you can see that the Engineering (material) Data and Geometry have already been defined (green check marks).
- It remains to set up and run the FE model in Mechanical
- Highlight the Engineering Data Cell and open by clicking on the Right Mouse Button (RMB)=>Edit to verify the linear material properties.
- Verify that the units are in Metric(Tonne,mm,...)
 - Utility Menu=>Units=>Metric(Tonne, mm,...)

Project Schematic

A		
1	Static Structural (ANSYS)	
2	Engineering Data	✓
3	Geometry	✓
4	Model	⚡
5	Setup	✓
6	Solution	✓
7	Results	✓

Surface Body Contact, Static Structural

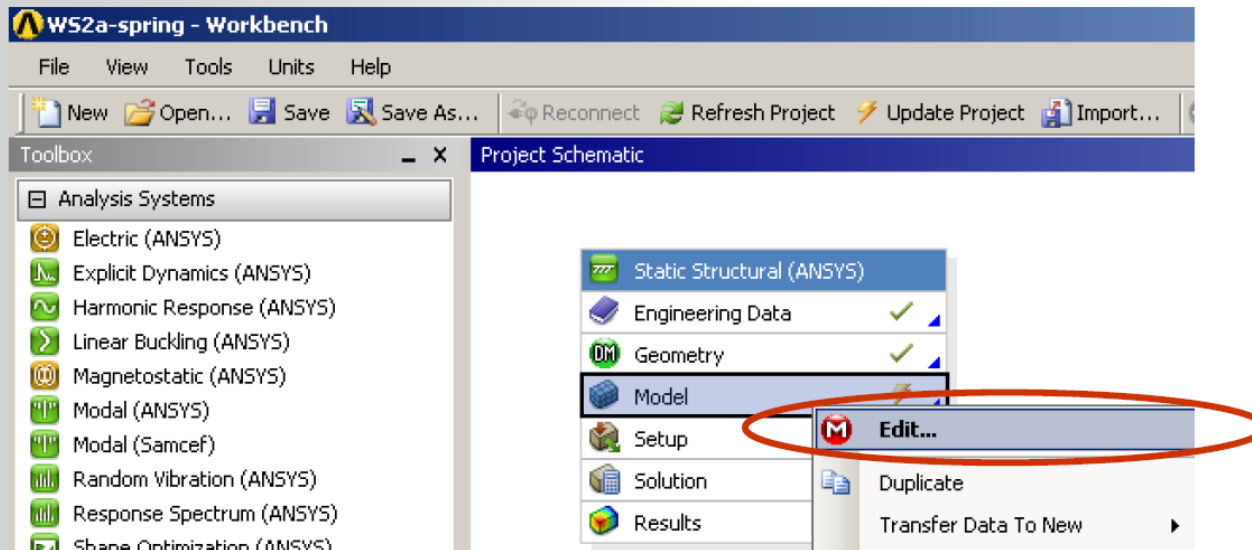
Properties of Outline Row 3: Structural Steel			
	A	B	
1	Property	Value	U
2	Density	7.85E-09	tonne
3	Coefficient of Thermal Expansion		
4	Coefficient of Thermal Expansion	1.2E-05	C ⁻¹
5	Reference Temperature	22	C
6	Isotropic Elasticity		
7	Young's Modulus	2E+05	MPa
8	Poisson's Ratio	0.3	
9	Alternating Stress Mean Stress	Tabular	
10	Scale	1	

...Workshop 6A: Diagnostics

Close Engineering Data to return to Project Schematic

- Utility Menu > Return to Project

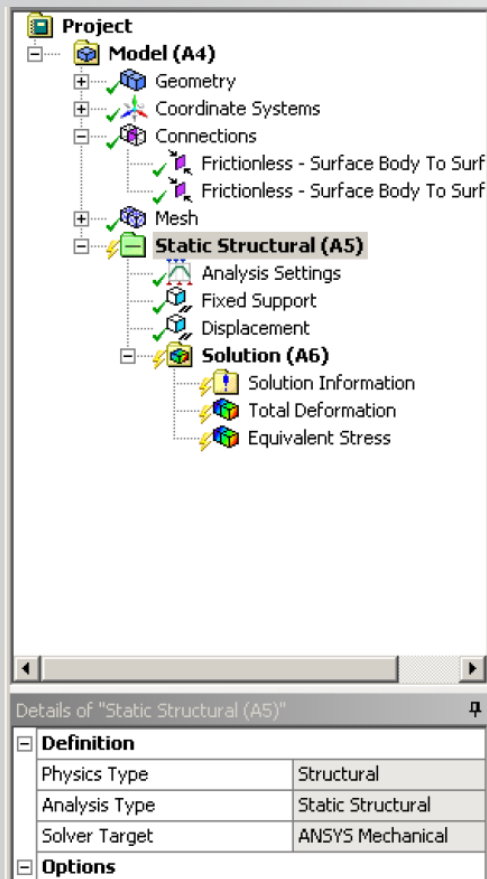
Double click on the Model Cell to open the FE Model (Mechanical Session)
(or RMB=>Edit...)



...Workshop 6A: Diagnostics

The spring assembly is already set up with frictionless contact pairs, a fixed boundary condition and a displacement load on the rigid component.

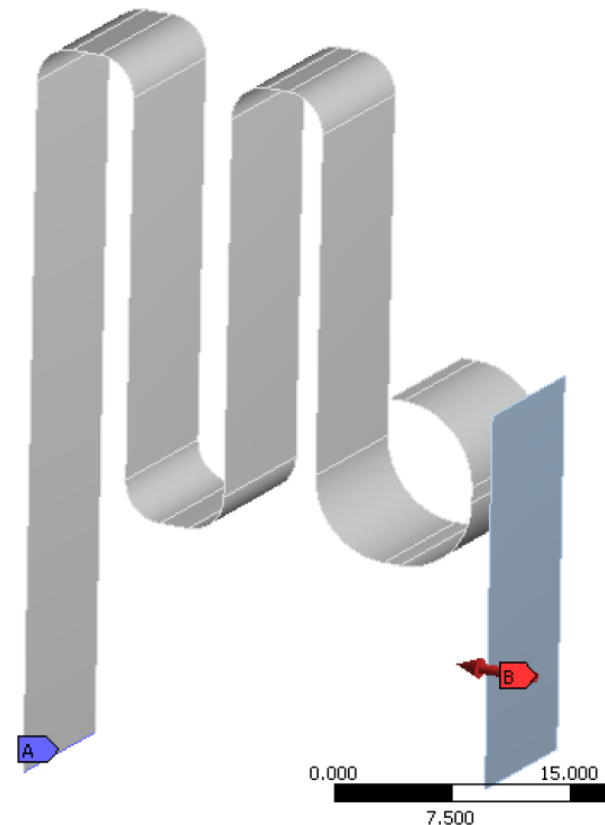
Highlight the entities beneath each folder to become familiar with the model and to confirm that it is properly supported and loaded and ready to solve.



A: Surface Body Contact, Static Structural

Static Structural
Time: 1. s
3/6/2009 9:51 AM

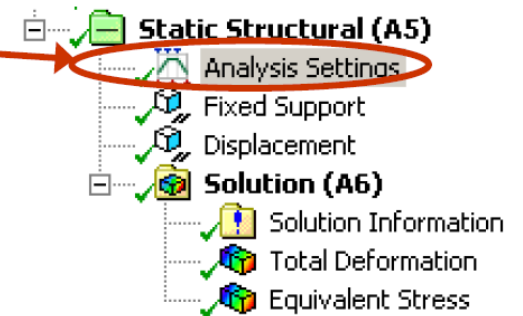
- A Fixed Support
- B Displacement



...Workshop 6A: Diagnostics

Confirm the Analysis Settings Specifications as shown:

Details of "Analysis Settings"	
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	On
Define By	Substeps
Initial Substeps	25.
Minimum Substeps	10.
Maximum Substeps	100.
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Large Deflection	On
Inertia Relief	Off



Run the Solution...



...Workshop 6A: Diagnostics

After solution run is complete, highlight the Solution Information folder and scroll to near the bottom of the output.

Note: You can also use Ctrl+F in solver output window to search file.

- The model solves without error.

The screenshot shows the ANSYS Workbench interface. In the left-hand tree view, the 'Solution (A6)' folder is expanded, and the 'Solution Information' sub-folder is highlighted. Below the tree view, a table titled 'Solution Information' provides details about the solver output.

Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Update Interval	2.5 s

```

DISP CONVERGENCE VALUE  = 0.7676E-13  CRITERION= 0.7653E-01 <<< CONVERGED
EQUIL ITER   1 COMPLETED.  NEW TRIANG MATRIX.  MAX DOF INC=  0.7676E-13
DISP CONVERGENCE VALUE  = 0.7676E-13  CRITERION= 0.7809E-01 <<< CONVERGED
LINE SEARCH PARAMETER =  1.000      SCALED MAX DOF INC =  0.7676E-13
FORCE CONVERGENCE VALUE = 0.1630E-07  CRITERION= 0.5206E-04 <<< CONVERGED
MOMENT CONVERGENCE VALUE = 0.1753E-08  CRITERION= 0.1041E-03 <<< CONVERGED
Writing NEWTON-RAPHSON residual forces to file: file.res003
>>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION   1
*** LOAD STEP   1  SUBSTEP  12  COMPLETED.    CUM ITER =   13
*** TIME =  1.00000      TIME INC =  0.700000E-01

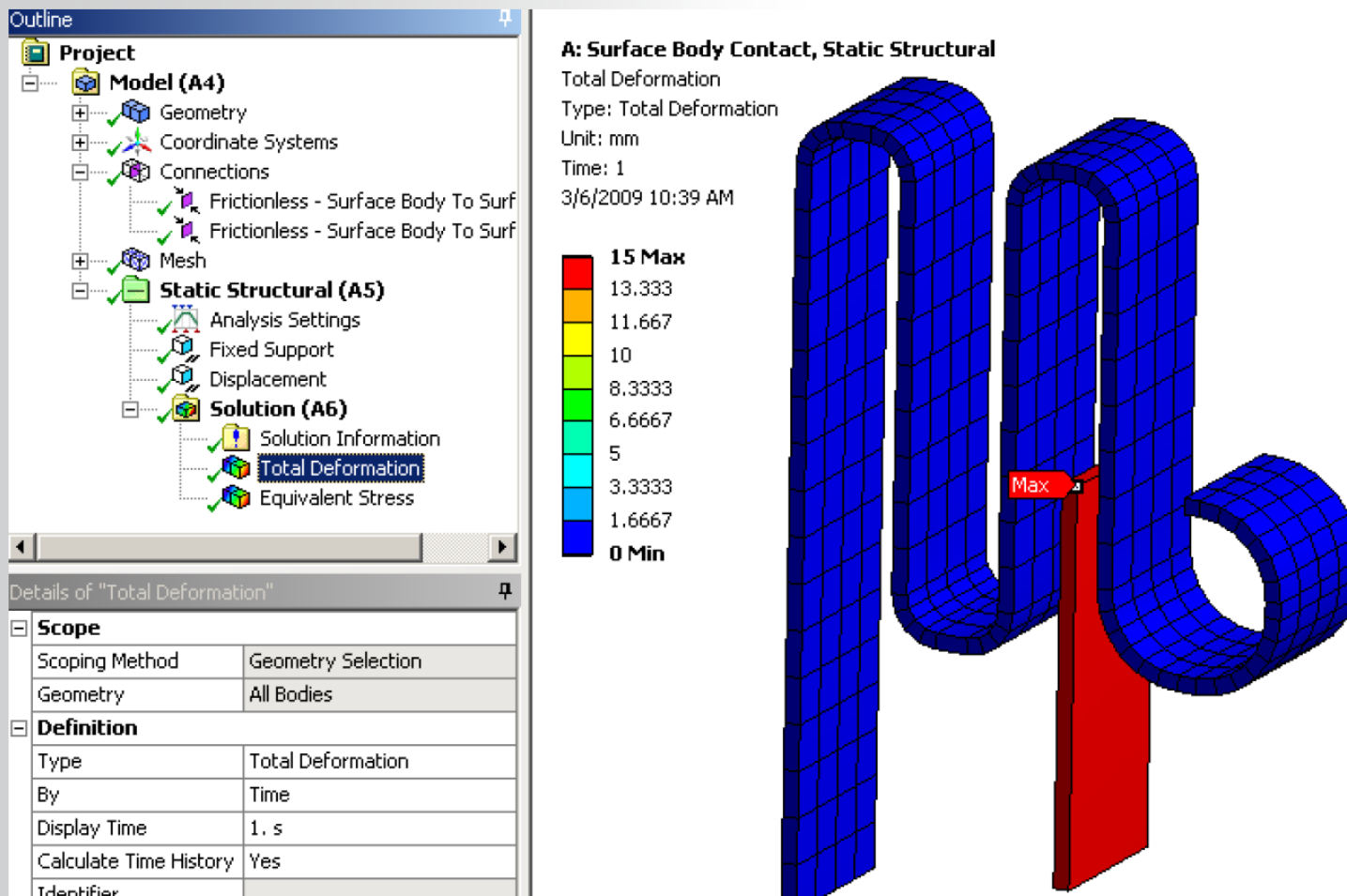
*** ANSYS BINARY FILE STATISTICS
BUFFER SIZE USED= 16384
    7.750 MB WRITTEN ON ELEMENT SAVED DATA FILE: file.esav
    0.688 MB WRITTEN ON ASSEMBLED MATRIX FILE: file.full
    10.250 MB WRITTEN ON RESULTS FILE: file.rst
***** Write FE CONNECTORS *****

WRITE OUT CONSTRAINT EQUATIONS TO FILE=
file.ce
  
```


...Workshop 6A: Diagnostics

Review the Total Deformation results.

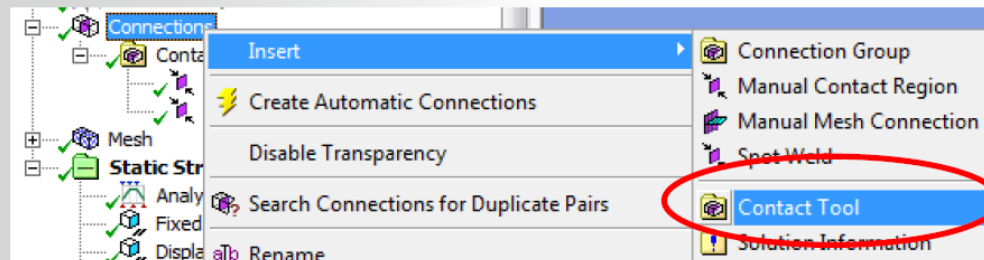
- Something is wrong. The contact relationship between the two parts has obviously failed.



...Workshop 6A: Diagnostics

With 25 initial substeps and Auto Time Stepping turned ON, the contact should have engaged. In an effort to determine the problem, we will evaluate what the initial condition of the contact pairs are.

- Highlight the Connections Folder: RMB> Insert> Contact Tool



- Highlight the Contact Tool: RMB>Initial Information>Generate Initial Contact Results. This will run a partial solve to establish initial contact parameters (i.e. Initial status, gap, penetration, etc...).



Review the Initial Contact Information.

Note the following:

- The two active pairs both have an initial status of “Far Open”
- Both pairs have a pinball radius of 4mm. Is 4mm enough?

Initial Information

For additional options, please visit the context menu for this table (right mouse button)

Name	Contact Side	Type	Status	Number Contacting	Penetration (mm)	Gap (mm)	Geometric Penetration (mm)	Geometric Gap (mm)	Resulting Pinball (mm)
Frictionless - Surface Body To Surface Body	Contact	Frictionless	Far Open	0.	0.	0.	0.	0.	4.
Frictionless - Surface Body To Surface Body	Target	Frictionless	Inactive	N/A	N/A	N/A	N/A	N/A	N/A
Frictionless - Surface Body To Surface Body	Contact	Frictionless	Far Open	0.	0.	0.	0.	0.	4.
Frictionless - Surface Body To Surface Body	Target	Frictionless	Inactive	N/A	N/A	N/A	N/A	N/A	N/A

Color Legend

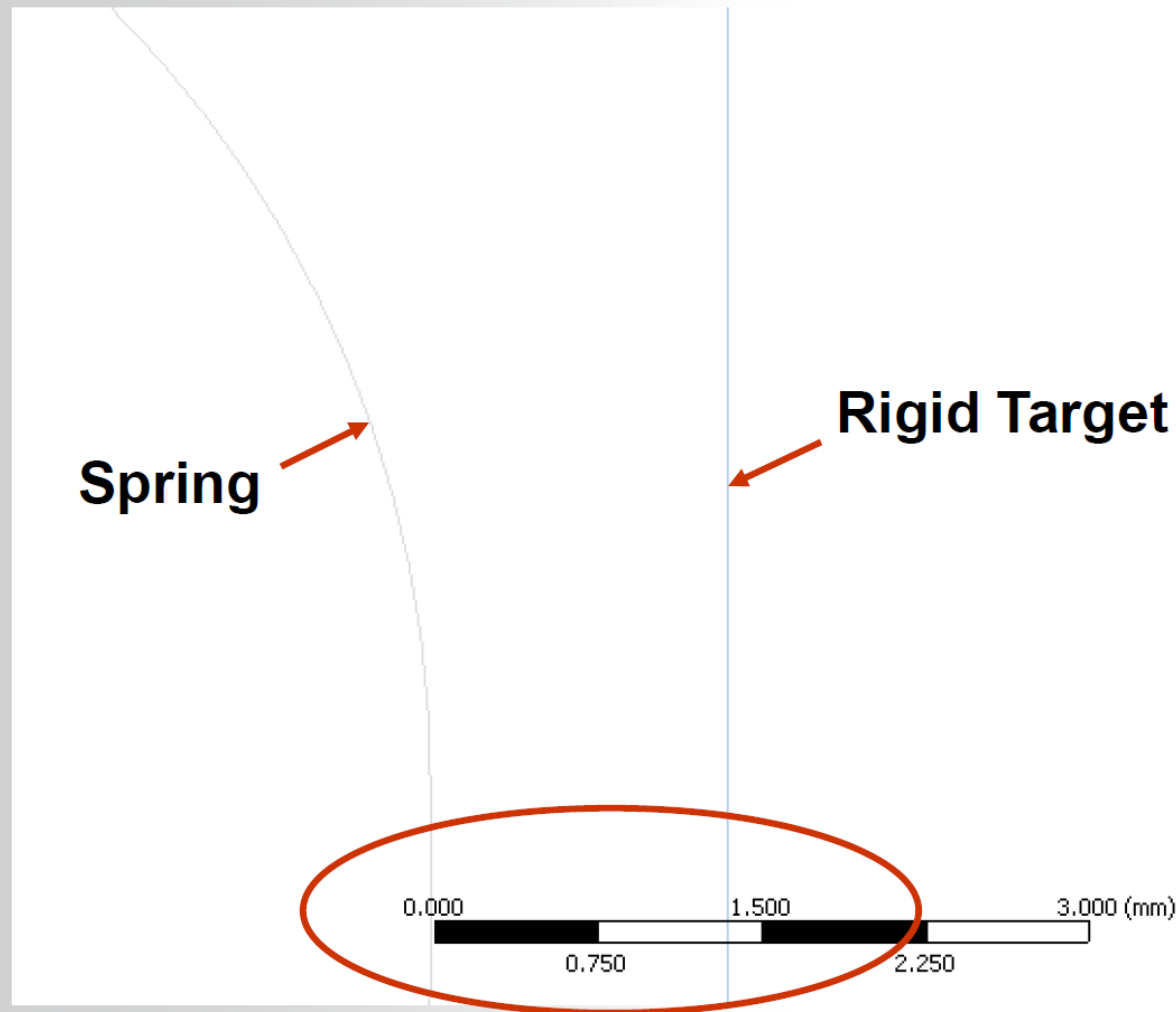
Red	The contact status is open but the type of contact is meant to be closed. This applies to bonded and no separation contact types.
Yellow	The contact status is open. This may be acceptable.
Orange	The contact status is closed but has a large amount of gap or penetration. Check penetration and gap compared to pinball and depth.
Gray	Contact is inactive. This can occur for MPC and Normal Lagrange formulations. It can also occur for auto asymmetric behavior.

For further details consult the online help at the following location:
 Mechanical Help > Using Mechanical Features > Results in Mechanical > Structural Results > Contact Tool

Geometry **Worksheet** / Print Preview / Report Preview

...Workshop 6A: Diagnostics

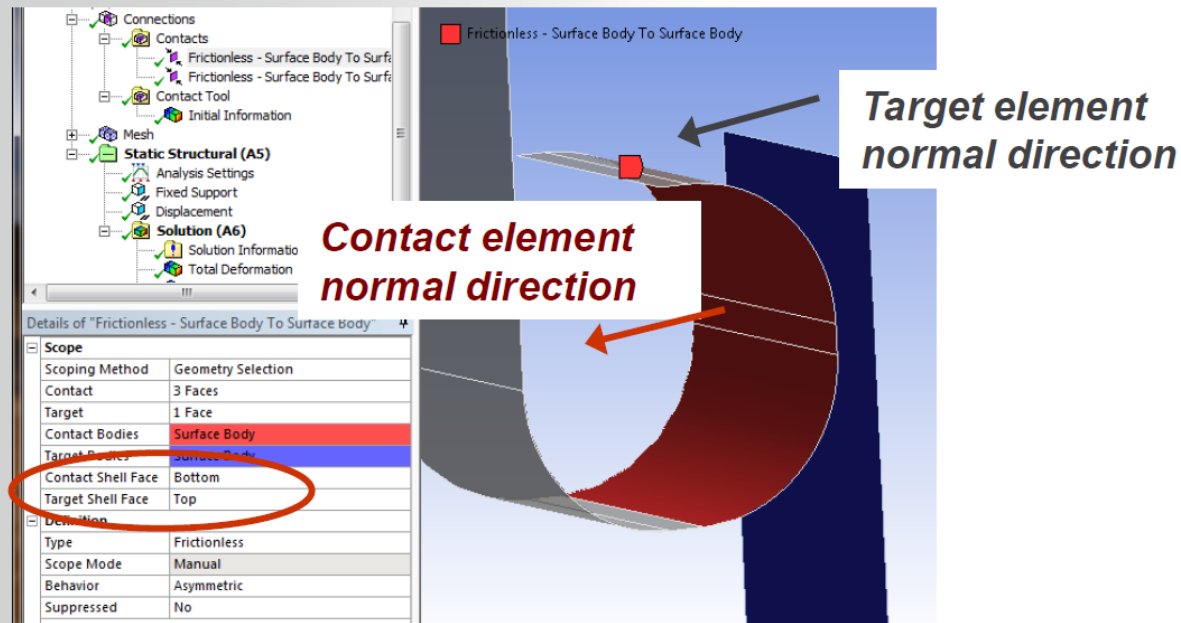
By studying a profile of the undeformed geometry we can see that the initial gap is less than 1.50mm. Hence, the Pinball Radius of 4mm should be sufficient for this contact pair to be in an initial status of 'near-open'.



...Workshop 6A: Diagnostics

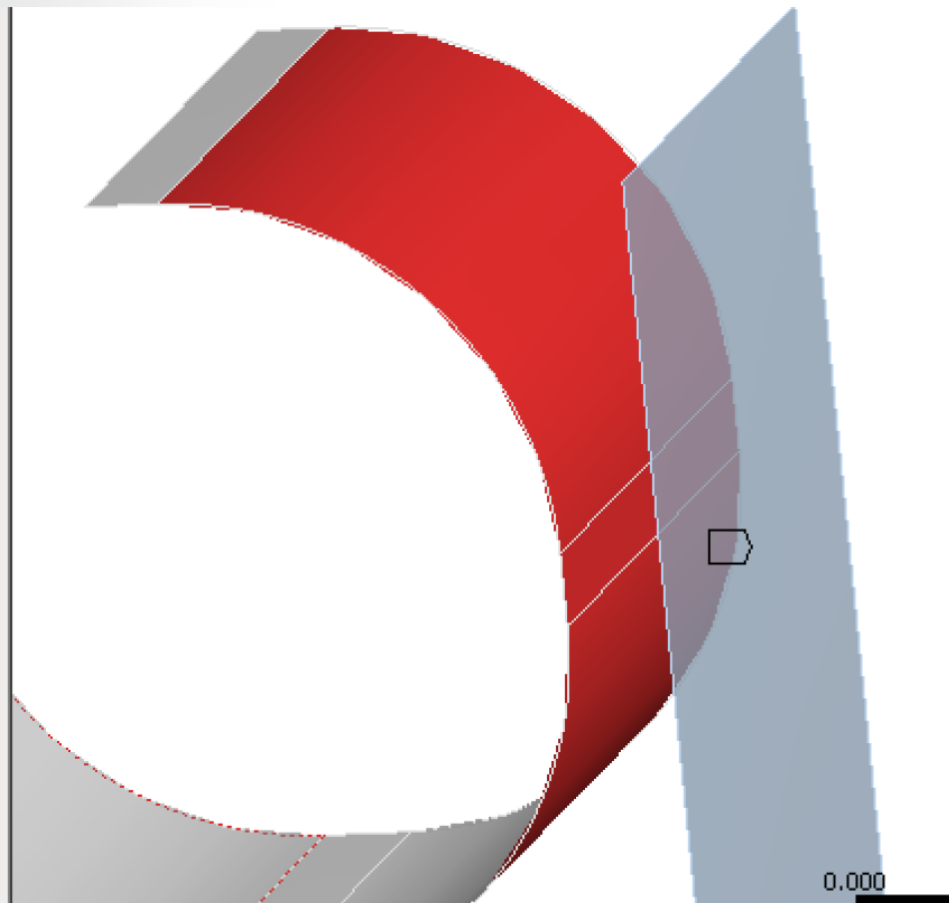
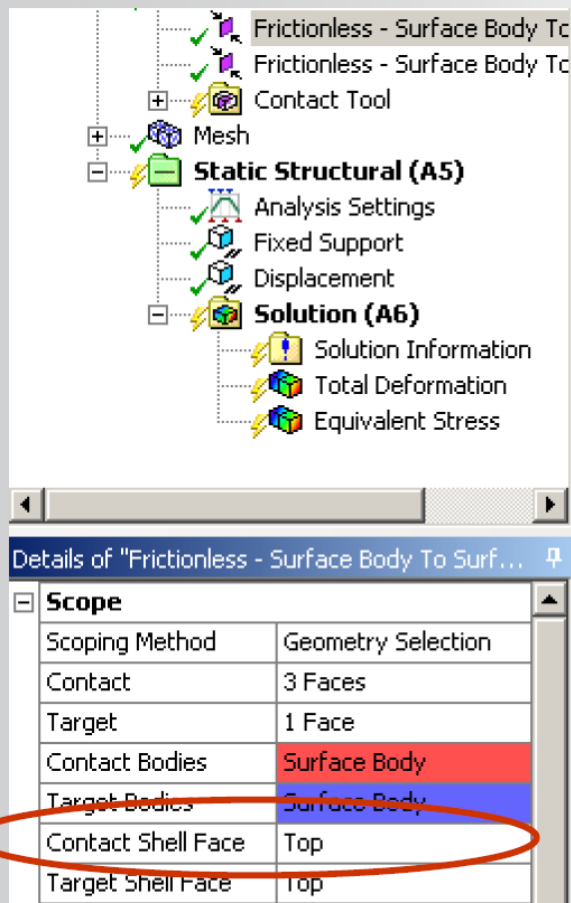
Highlight the contact region representing the contact between the spring and target.

- In order for contact to work properly, the contact element normals must be facing the target element normals.
- Recall that surface bodies are meshed with shell elements that have a 'top' and a 'bottom' face. The reason this contact pair is not working is because the contact normals are on the wrong side of the surface body with normals that face away from the target. This needs to be reversed.



...Workshop 6A: Diagnostics

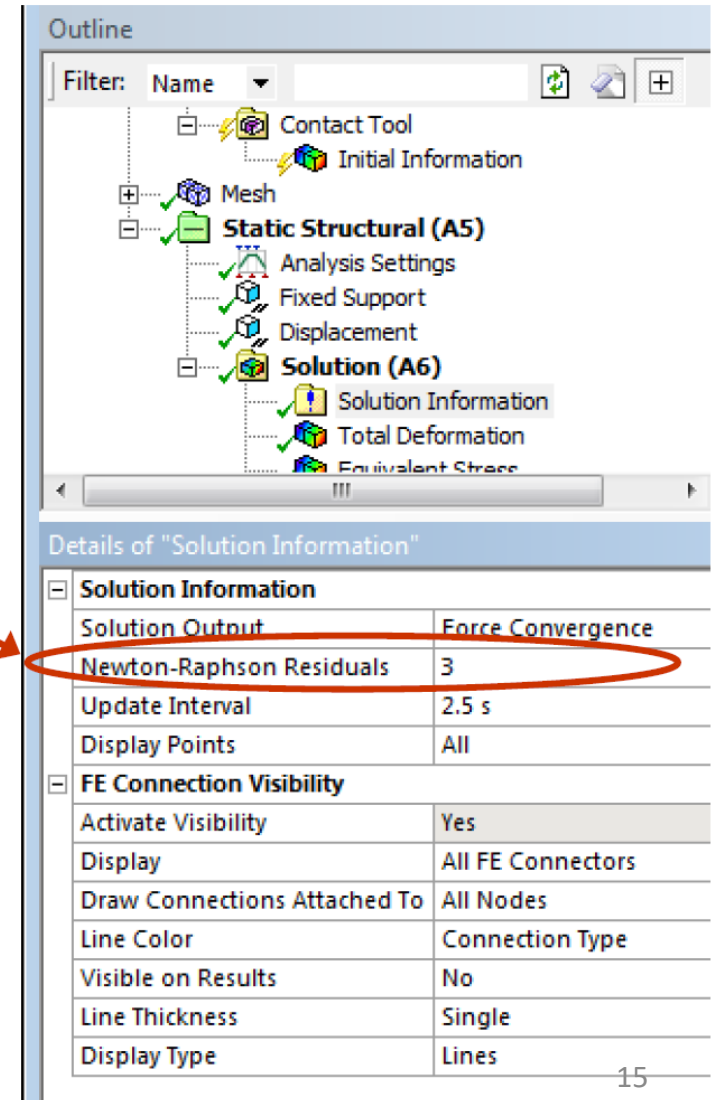
From the details window of the contact region, switch the 'contact shell face' from Bottom to Top . The red color highlighting the contact side should switch.



Highlight the Solution Information Branch

- Set Newton-Raphson Residuals = 3
- This will save force imbalance data for the last '3' Newton-Raphson iterations. This is especially helpful information for troubleshooting troubling contact convergence problems.

Rerun the solution



Outline

Filter: Name

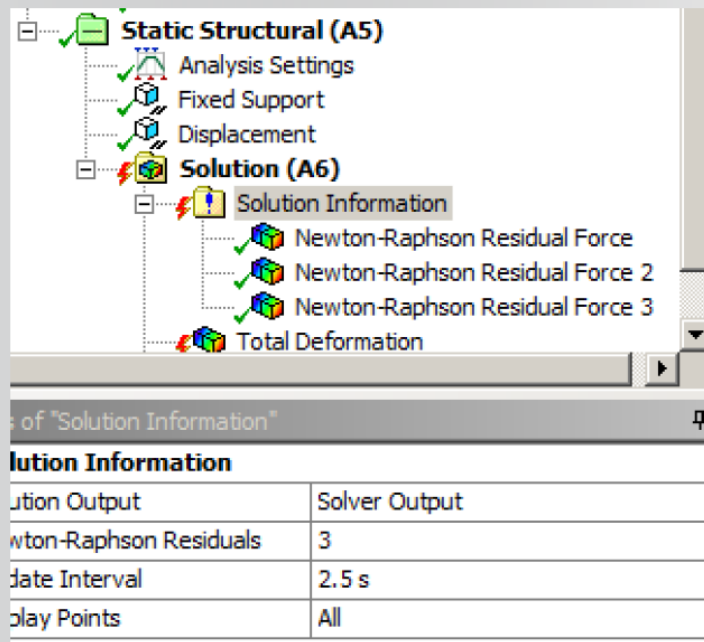
- Contact Tool
- Initial Information
- Mesh
- Static Structural (A5)
 - Analysis Settings
 - Fixed Support
 - Displacement
 - Solution (A6)
 - Solution Information
 - Total Deformation
 - Equivalent Stress

Details of "Solution Information"

Solution Information	
Solution Output	Force Convergence
Newton-Raphson Residuals	3
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

...Workshop 6A: Diagnostics

From the Solution Information Branch, the contact is now engaging, but the solution fails to converge after several iterations.



Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	3
Time Interval	2.5 s
Display Points	All

```

LINE SEARCH PARAMETER = 0.5000E-01 SCALED MAX DOF INC = 0.8193E-02
FORCE CONVERGENCE VALUE = 188.6 CRITERION= 2.685
MOMENT CONVERGENCE VALUE = 0.6505E-02 CRITERION= 0.7611E-01 <<< CONVERGED
Writing NEWTON-RAPHSON residual forces to file: file.nr003
  
```

```

*** ERROR *** CP = 34.492 TIME= 14:44:27
Solution not converged at time 0.1 (load step 1 substep 4).
Run terminated.
  
```

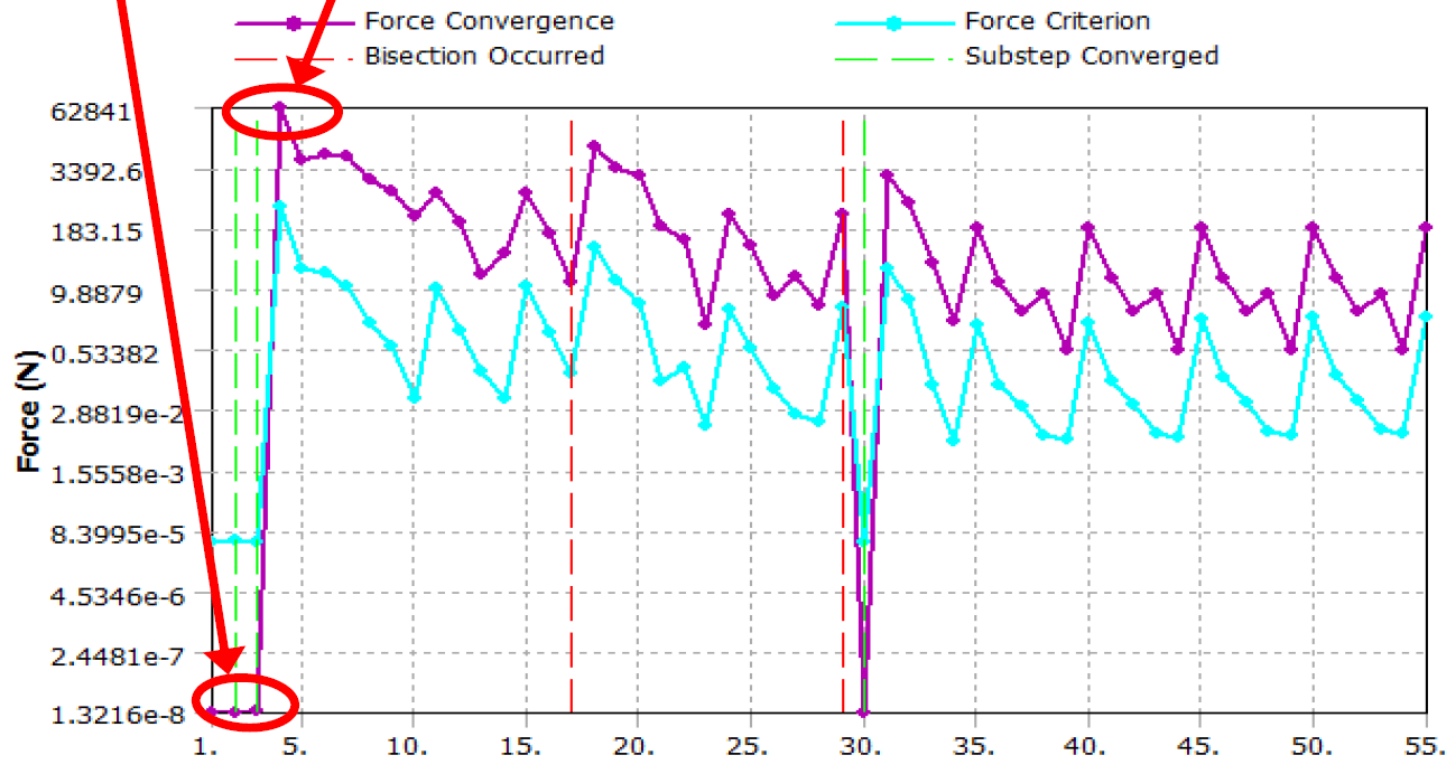
```

*****
SUMMARY FOR CONTACT PAIR IDENTIFIED BY REAL CONSTANT SET 3
Max. Penetration of -6.881037057E-03 has been detected between contact
element 389 and target element 414.
For total 52 contact elements, there are 4 elements are in contact.
Contacting area 10.0000034.
Contact element 389 has the highest number of status changes (total 11)
during this substep.
Max. Total sliding distance 8.099771024E-02.
Max. Pinball distance 4.
One of the contact searching regions contains at least 1 target
  
```

...Workshop 6A: Diagnostics

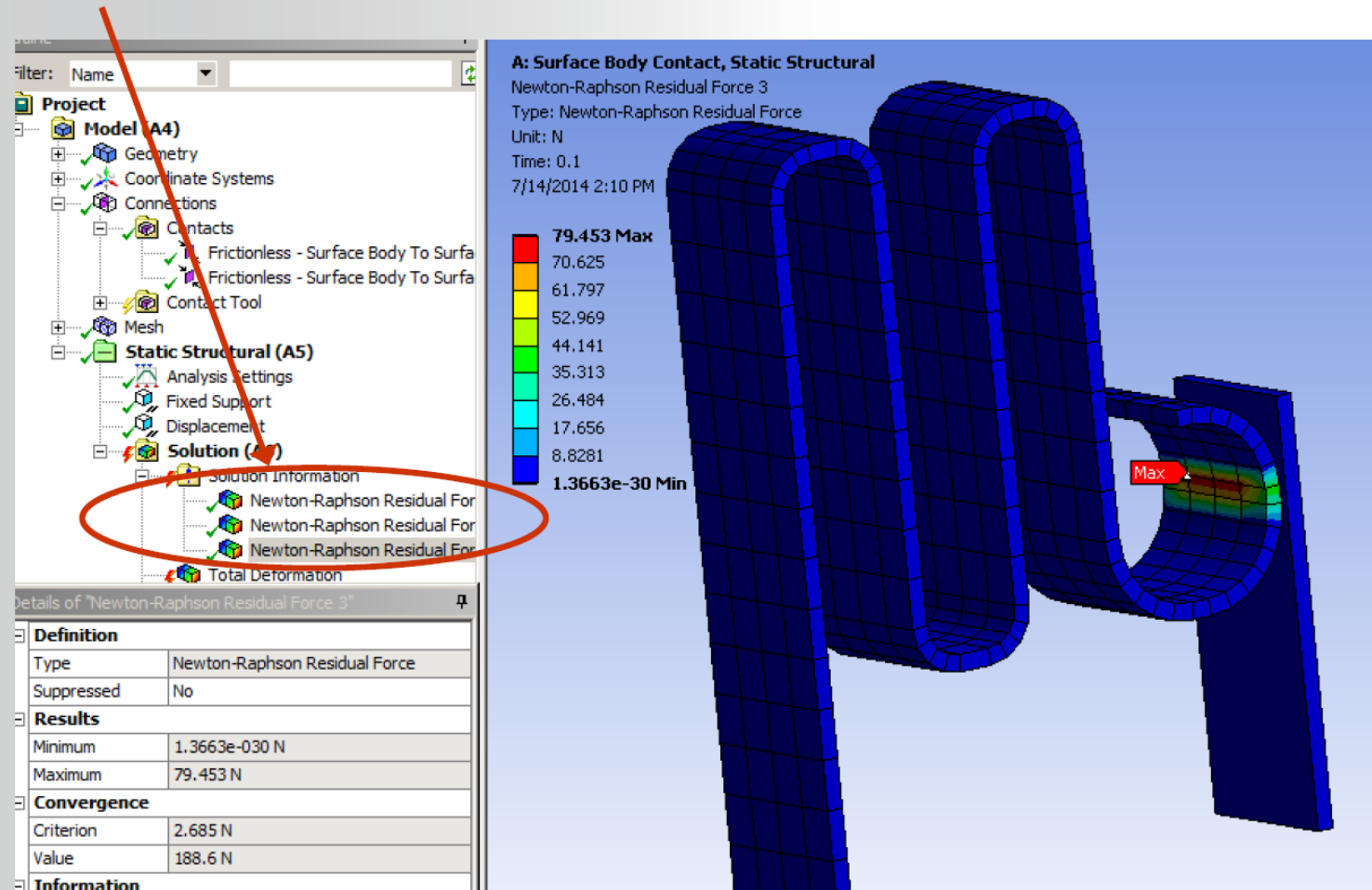
These first two converged substeps likely represent the trivial solutions that occur as the small gap between the two parts is being closed and no contact has been made yet.

This first spike in the Newton-Raphson residual (measure of imbalance) likely occurs at the point when contact first engages. From there on out, the solution struggles and fails after two bisections and many iterations to find a balance.



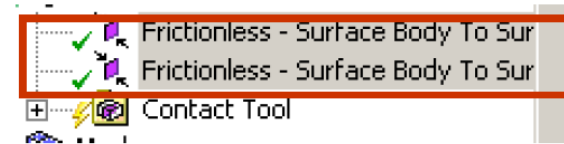
...Workshop 6A: Diagnostics

A plot of Newton-Raphson Residual (measure of force imbalance in the model) confirms that the point where contact is engaged is the source of the highest imbalance .



...Workshop 6A: Diagnostics

- Highlight both Contact Regions and change the contact specifications in the details window.



Reducing the contact stiffness factor will reduce the calculated force generated at the contact surface and thereby reduce the imbalance

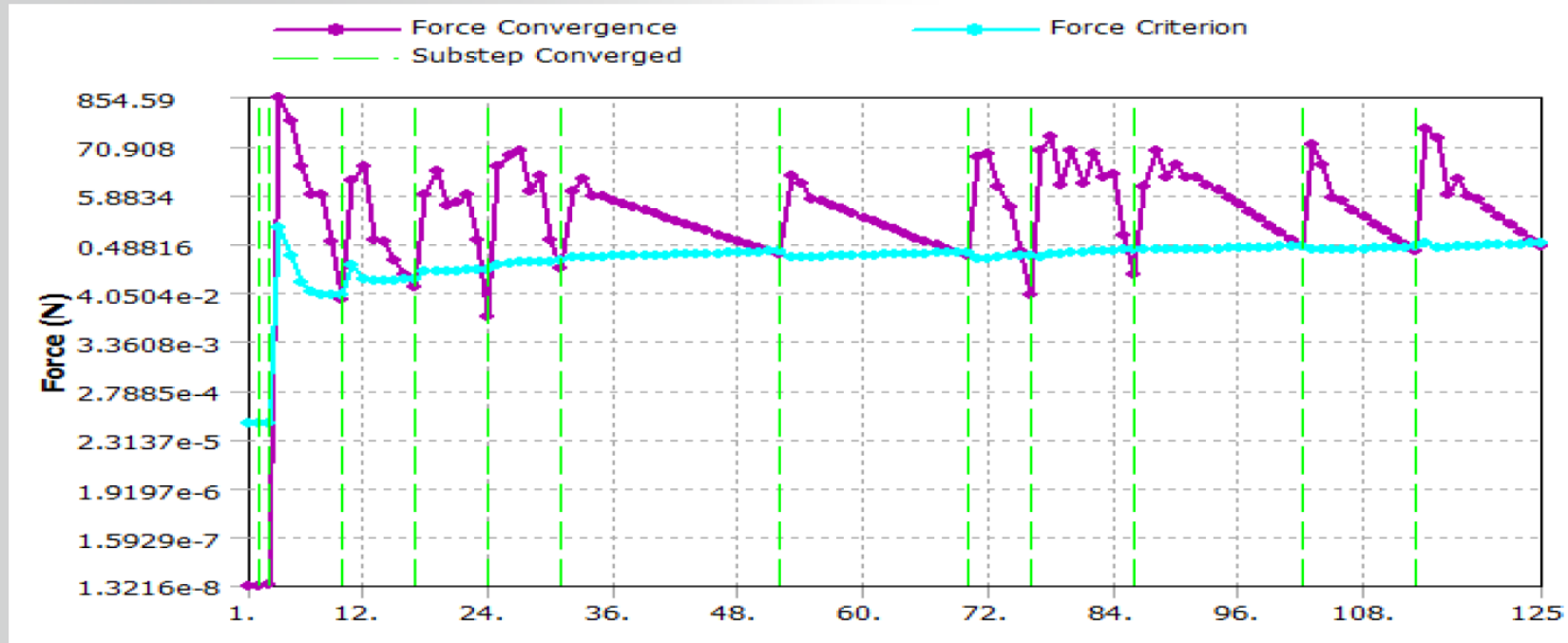
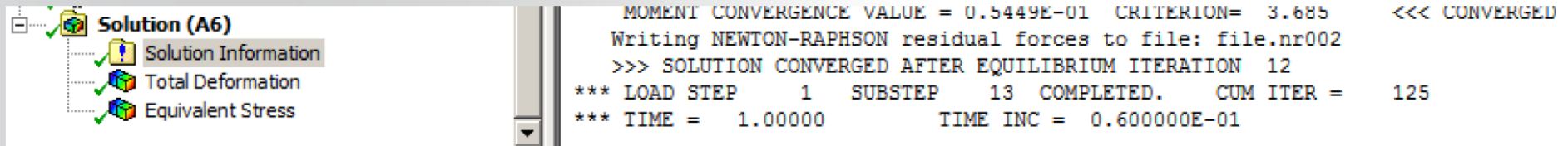


[-] Scope	
Scoping Method	Geometry Selection
Contact Shell Face	Top
Target Shell Face	Top
Shell Thickness Effect	No
[-] Definition	
Type	Frictionless
Scope Mode	Manual
Behavior	Asymmetric
Trim Contact	Program Controlled
Suppressed	No
[-] Advanced	
Formulation	Program Controlled
Detection Method	Program Controlled
Penetration Tolerance	Program Controlled
Interface Treatment	Add Offset, No Ramping
☐ Offset	0. mm
Normal Stiffness	Manual
Normal Stiffness Factor	1.e-002
Update Stiffness	Program Controlled
Stabilization Damping Factor	0.
Pinball Region	Program Controlled
Time Step Controls	None

- Re-run the solution

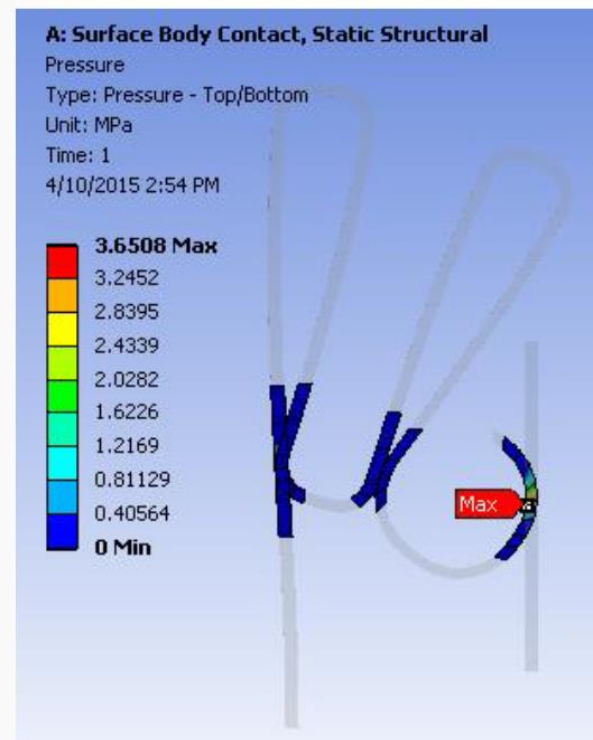
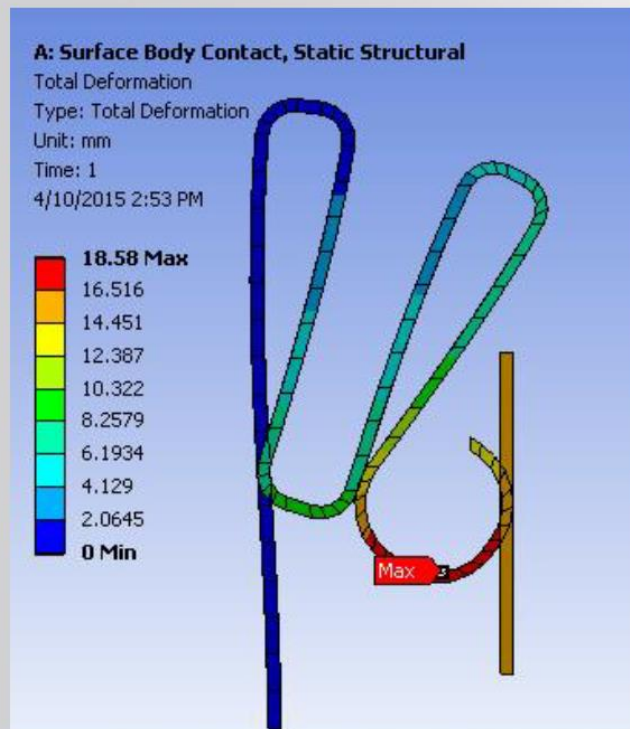
...Workshop 6A: Diagnostics

The solution now converges very nicely with no bisections



Review the Total Deformation results.

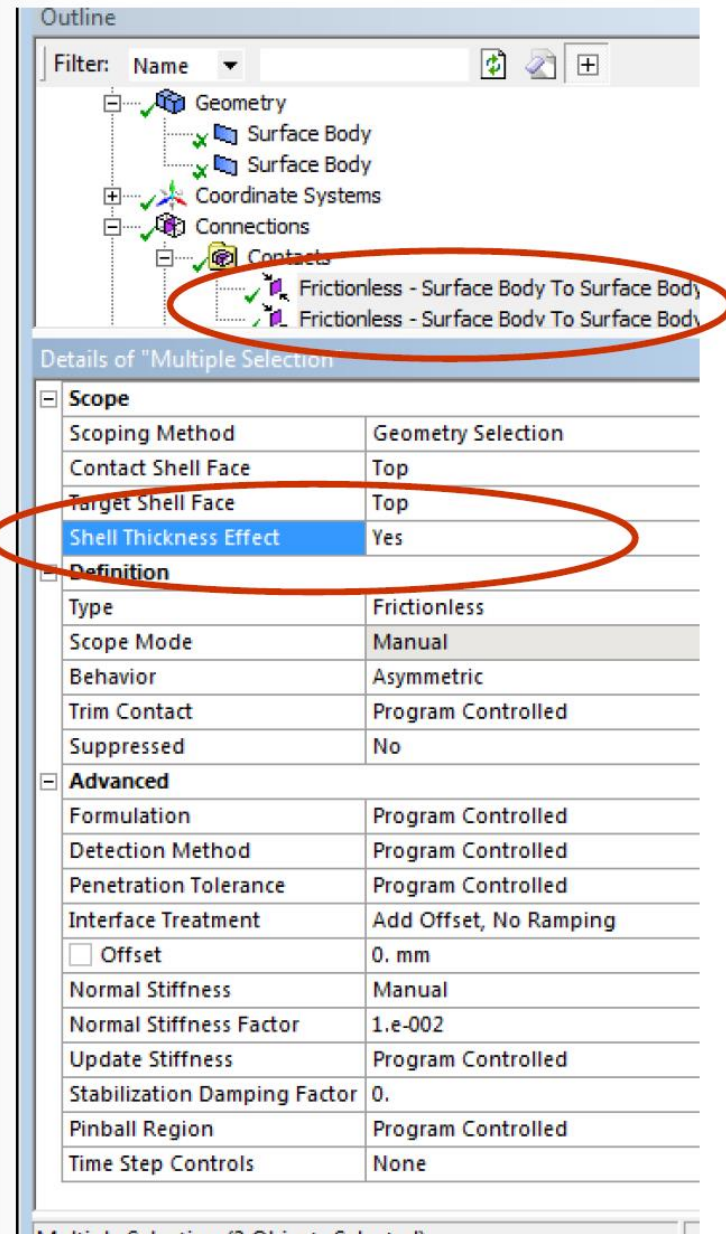
- Although this solution is now converged, notice the excessive penetration. This is because, by default the shell contact detection points are at the midplane of the shells.



- By default, contact elements are located at the mid plane of surface bodies, in same position as the shell elements that represent these bodies.

...Workshop 6A: Diagnostics

- Highlight the Contact Regions and return to the Details window
 - Set “Shell Thickness Effect” to “Yes”
- Re-run the solution



...Workshop 6A: Diagnostics

Review the Total Deformation results as before.

- Shell thickness is now properly accounted for.

